

FileUpload Application

A basic, back-end file upload application built with FastAPI backend, featuring azure keyvault, azure storage containers, and Docker containerization.

(Plan to include frontend, database, user registration, login, etc in the future)

Features

- **File Upload:** Upload a file to an azure storage container
- **List Existing Containers:** Fetch list of existing storage containers
- **Create Containers:** Create a new storage container
- **Containerized:** Full Docker setup for easy deployment

Technology Stack

Backend (/)

- **FastAPI:** Modern, fast web framework for APIs

Project Structure

```
file_upload/
├── py-file-upload/           # Python FastAPI backend
│   ├── main.py               # FastAPI application connection
│   ├── requirements.txt      # Python dependencies
│   ├── .env                  # Environment variables
│   └── Dockerfile            # API container config
├── rt-file-upload/          # React frontend
│   ├── src/
│   │   ├── components/      # React components
│   │   ├── context/         # React context providers
│   │   ├── services/        # API service layer
│   │   └── index.js          # App entry point
│   ├── public/              # Static assets
│   └── package.json          # Node.js dependencies
```

Quick Start

Prerequisites

- Docker
- Git
- Azure Subscription
- Azure App Registration
- Azure Keyvault
- Azure Storage Account

- Azure Storage Account Container

Installation and Setup

1. Clone and navigate to the project:

```
cd file_upload
```

2. Start all services:

```
docker build -t py-file-upload-image .  
docker run -d --name py-file-upload-container -p 8000:8000 py-file-upload-  
image:latest
```

3. Access the application:

- API: <http://localhost:8000>
- API Documentation: <http://localhost:8000/docs>

Manual Setup (Development)

Backend Setup

```
cd api  
pip install -r requirements.txt  
py -3.13 -m uvicorn main:app --host 0.0.0.0 --port 8000
```

API Endpoints

File upload

- **GET** [/containers/](#) - List all storage containers
- **POST** [/containers/create/](#) - Create a storage containers
- **POST** [/containers/uploadfile/](#) - Upload a file to the container
- **GET** [/containers/blobs/](#) - List all files of a container

Usage

1. List all available containers
2. Pick on container
3. Upload a file to that container
4. Check the container by listing all files

Environment Variables

Backend (.env)

```
ORIGIN=<frontend_url>  
KEY_VAULT_NAME=<keyvault_name>  
STORAGE_ACCOUNT_NAME=<storage_account_name>  
AZURE_CLIENT_ID=<appId>  
AZURE_TENANT_ID=<tenantId>  
AZURE_CLIENT_SECRET=<password>
```

Contributing

1. Fork the repository
2. Create a feature branch
3. Make your changes
4. Test thoroughly
5. Submit a pull request

License

This project is open source and available under the MIT License.